Group 10 - Section 2

Implementation of a PSO-Based Security Defense Mechanism for Tracing the Sources of DDoS Attacks

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PSO-Based Security Defense for Tracing DDoS Sources

- Addresses how to trace back the real sources of distributed denial-of-service (DDoS) attacks
- Existing work focuses on firewalls and intrusion detection systems, not on reconstructing the exact network routes taken by malicious packets.
- Concentrates on the IP traceback (IPTBK) problem
- Key idea is to use a modified Particle Swarm Optimization (PSO) algorithm together with a hash-based IP traceback mechanism called SPIE (Source Path Isolation Engine)
- To efficiently reconstruct the most probable attack routes

Project Objective and Scope

Core Objective:

- Design a PSO-based IP traceback model (PSO-IPTBK) that reconstructs the most probable routes used by DDoS attack traffic between attackers and a victim server

Key Tasks:

- Statistical analysis of traffic behavior to estimate which routers and links carried the bulk of the attack traffic
- Binary classification trough the identification of abnormal patterns: normal vs. attack traffic
- Detect which hosts are most likely to be attack sources (zombies)
- Decide a small set of high-probability attack paths from victim back to attackers.

Project Objective and Scope

Scope of Study:

- Simulated networks with 24 nodes and 40 nodes using OMNeT++ and the INET framework.
- Focus on UDP-based DDoS scenarios using compromised IoT devices.

Out of Scope:

- Full attack prevention or mitigation mechanisms.
- This project focuses solely on detection.

Project System Overview

- Traffic Monitoring
  →Continuous packet capture and measurement

- Feature Extraction
  → Calculate packet counts and rates

- Probabilistic Modeling
  → Apply statistical distributions

- Decision Making
  → Evaluate probability thresholds

- Alert Generation
  → Trigger attack notifications

Sources of Uncertainty

Network traffic analysis inherently involves uncertainty from multiple sources. Understanding these uncertainties is critical for building robust detection systems.

Measurement Uncertainity:

- Packet sampling errors in monitoring systems
- Noise from network measurement tools
- Timing inaccuracies in data collection

Environmental Variability:

- Changing user behavior patterns
- Time-dependent traffic fluctuations
- Seasonal and periodic variations

Modeling Uncertainty:

- Simplified assumptions about traffic
- Approximation of real-world complexity
- Limited model expressiveness

Key Random Variables and Their Roles

Primary Variable:

- Packet arrival count per time window: The number of packets observed in fixed time intervals serves as the fundamental random variable for detection.

Role in Detection:

- These variables quantify normal traffic behavior and identify statistically significant deviations that indicate potential DDoS attacks.

Derived Variables:

- Average traffic rate: Long-term mean of packet arrivals
- Traffic deviation: Difference between observed and expected rates
- Probability density: Likelihood of current traffic pattern

Probabilistic Models and Assumptions

Model Selection: Implicit Probabilistic Modeling

Poisson distribution models packet arrival counts, widely used in network traffic analysis for its simplicity and interpretability.

Key Assumptions:

- Independent packet arrivals
- Constant average rate in short intervals
- Stationary traffic characteristics
- Sampling model: ~3% independent packet logging.
- Higher packet counts imply higher attack path probability.
- PSO uses stochastic search model for velocity updates (rand()).
- Packets conditionally independent given a route.
- Fitness and coverage act as probability proxies.

Probabilistic Reasoning and Inference Logic

Steps for Analyzing Traffic Patterns:

- Model Normal Behavior
  - Estimate anticipated traffic patterns based on historical data.

- Observe Current Traffic
  - Measure real-time packet arrival rates.

- Compute Probability
  - Calculate the likelihood of the observed traffic according to the normal model.

- Compare Threshold
  - Assess the measurements against the decision threshold.

Operationalizing Probabilistic Reasoning

Evaluation Framework for Traffic Detection

- Monte Carlo Simulation
  - Create synthetic traffic scenarios to evaluate detection performance in a controlled setting.
- Traffic Simulation
  - Model both standard traffic patterns and DDoS attack scenarios with different intensities.
- Performance Metrics
  - Assess the detection rate (true positives) alongside the probability of false alarms (false positives).
- Assumptions
  - Fixed time windows and stationary traffic characteristics are taken into account within brief observation periods.

Current Limitations and Conceptual Gaps

Known Model Limitations:

- Poisson model may miss bursty traffic patterns
- Fixed thresholds can cause false alarms
- Limited handling of traffic variability

Conceptual Gaps:

- No distinction between flash crowds and attacks
- Limited adaptability to traffic changes
- Stationarity assumption may not hold

Planned Refinements and Coordination

Technical directions:

- Test on larger, more realistic topologies.
- Make fitness more explicitly probabilistic (likelihood-based).
- Explore other swarm / evolutionary variants.

Conceptual work:

- Better model sampling noise, routing changes, benign traffic.
- Integrate other traceback schemes (PPM, DPM, ACO, etc.).

Coordination:

- Reviewer/manager to track assumptions, experiments, code and comparisons.

Summary of Group-Level Understanding

- The paper discusses DDoS attacks using compromised IoT devices to flood a victim with traffic.
- Spoofed IPs lead to the IP traceback problem.
- It uses a Particle Swarm Optimization (PSO) algorithm for probabilistic search.
- PSO particles represent potential attack routes.
- Routers log sampled packet digests with SPIE support.
- PSO identifies likely routes based on traffic.
- Simulations show high accuracy in tracing attack paths.
- Smaller networks perform better, but scaling is challenging.
- Swarm intelligence effectively traces DDoS attacks with incomplete data.

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Project Context and Motivation

Domain: Network Security & Cyber Defense

The Core Problem: DDoS attackers use Spoofed IPs to hide their location.

We cannot see the whole picture because routers only sample 3% of packets to save memory.

Why is Uncertainty Intrinsic?

- Since we miss 97% of the data, we can never be 100% sure of the attack path.
- We must use probabilistic inference to estimate the most likely route based on incomplete evidence.

Project Objective and Scope

Exact Objective: To identify the most probable attack path from the attacker to the victim using incomplete packet data.

Methodology: We use Particle Swarm Optimization (PSO). The AI agents (particles) search through thousands of potential routes to find the one that best matches the 3% packet evidence.

Decision Output: The system decides on a specific sequence of routers that maximizes the probability:

$$P(\text{Route } R \mid \text{Observed Packets})$$

System Overview (SPIE & PSO)

1. Data Collection (SPIE):

- Routers independently sample packets using a Bernoulli trial (p=0.03).
- Logged packet digests are sent to the traceback server.

2. Processing Stage (PSO Algorithm):

- Initialization: Particles are placed randomly in the network.
- Search Loop: Particles move through the topology using probabilistic velocity vectors.
- Fitness Evaluation: Routes are scored based on how many sampled packets lie on that path.

Primary Sources of Uncertainty

Measurement Uncertainty (Packet Sampling):

- The primary source of randomness. We do not possess a complete record of traffic; the missing 97% creates blind spots.

Algorithmic Uncertainty (Stochastic Search):

- The PSO algorithm uses random numbers (r_1, r_2) to explore the network.
- Different runs might initially explore different paths before converging.

Environmental Variability:

- Background network noise fluctuates, meaning the baseline traffic varies over time.

Key Random Variables

S (Sampling Decision):

- Type: Bernoulli Random Variable (p=0.03).
- Role: Determines if a specific packet is recorded (1) or lost (0).

N (Total Packets Sampled):

- Type: Binomial Random Variable (n=1800, p=0.03).
- Role: Represents total evidence. Expected value E[N] = 54.

X_i (Particle Position):

- Type: Multivariate Random Variable.
- Role: Represents a candidate attack route being evaluated.

Probabilistic Models & Assumptions

Distributions Used:

- Bernoulli(0.03): Models individual router behavior.
- Uniform(0, 1): Models the exploration logic of the AI agent (r_1, r_2).

Key Assumptions:

1. Independence: Packet sampling at Router A is independent of Router B.
2. Acyclic Graph: We assume the attack path does not contain loops to simplify the probability space.
3. Representative Sample: the 3% observed packets accurately reflect the distribution of the full attack traffic.

Probabilistic Reasoning Logic

Bayesian Reasoning:

- We calculate likelihood: A route containing more observed packets is statistically more likely to be the true path.

Propagation of Uncertainty:

- Uncertainty starts at the Router (Bernoulli) → Dataset (Binomial) → Solver (PSO Fitness).

Dependency Chain:

- Particle movement is dependent on the Global Best (P_gbest). If the swarm finds strong evidence on one path, all particles are probabilistically pulled toward it.

Operationalizing Math into Code

From Math to Code:

- Bernoulli Variable: Implemented by generating a random number for every packet: if random() < 0.03: log_packet().
- Stochastic Velocity: Implemented using two independent rand() calls per iteration in the velocity update formula.

Evaluation Metric:

- Coverage Percentage: (Avg packets per hop / Total packets).
- Result: Routes with ~31-33% coverage were identified as the high-probability attack paths in our experiments.

Current Limitations & Gaps

Scalability Issue:

- Accuracy is 98.33% on 24 nodes but drops to 94.64% on 40 nodes. We lack a mathematical formula to predict this decay.

Sampling Rate Blind Spot:

- We use fixed p=0.03. We do not yet know the minimum p required to maintain 95% accuracy.

Idealized Conditions:

- The model assumes perfect router cooperation. We have not yet modeled non-cooperative routers or random packet loss due to congestion.

Planned Refinements & Summary

Milestone 3 Goals (Theoretical Guarantees):

- Scalability: Derive bounds for convergence probability as network size increases.
- Confidence Intervals: Add error bars to predictions
- Swarm Dynamics: Analyze the joint distribution of particles.

Summary:

- We successfully translated Traceback into a Probabilistic Model using Bernoulli/Binomial distributions.
- Conclusion: Traceback is about maximizing the Likelihood Function generated by the PSO algorithm.